RESINS
AND
RESIN COMBINATIONS

Topic outline

- General Properties and chemistry
- Examples of drugs containing resins:
  - Rosin
  - Podophyllin
  - Jalap
  - Mastic
  - Cannabis

  (Preparation, constituents, factors affecting the narcotic activity, Legal aspects, Analysis)
  - Oleoresins
  - Oleo-gum resins
  - Balsams.

Occurrence

- These are the amorphous mixtures of essential oils, oxygenated products of terpene and carboxylic acids found as exudations from the trunk of various trees.
- Resins are produced normally during growth or secreted as a result of injury to the plant.
- Formed in schizogenous or schizolysigenous ducts or cavities
- End products of metabolism and considered the oxidation product of terpenes

GENERAL PROPERTIES OF RESINS

- The term 'resin' is applied to more or less solid, amorphous products of complex chemical nature.
- They are insoluble in water and usually insoluble in petroleum ether but dissolve more or less completely in alcohol, chloroform and ether.
- solid or semisolid plants plant exudates produced normally (physiologically) or pathologically
- On evaporation of resin solution varnish like film is deposited.

General properties of resins

- Produce smoky flame on burning.
- On heating they soften and finally melt.
- Hard, transparent or translucent brittle materials.
- Amorphous materials but rarely crystallisable in nature.
- Resins are bad conductors of electricity.
- Specific gravity ranging from 0.9-1.25.
- Physically, resins are hard, transparent or translucent and heavier than water.
- Purified resins are amorphous and brittle

➢ Solubility

- Resins are practically
  - ➢ insoluble in highly polar solvents (water) and highly non-polar solvents (pet. ether) but
  - ➢ dissolve in organic solvents like alcohol, chloroform, and ether.
- On evaporation of resin solution varnish like film is deposited.

Why plants produce resins?

- To seal plant wounds
- To kill insects and fungi
- To eliminate excess metabolites
- Resins are obtained from the plant by :-

1. Incision in the bark or wood of the secreting plant(balsam of peru, balsam of tolu and benzoin)
2. Solvent extraction extracting the plant material with alcohol and precipitating the resin with water e.g jalap, podophyllum
3. Distillation from oleo resins, e.g rosin from turpentine

Classification

- On the basis of their occurrence in combination with other compounds, resins are classified into:
  a) Oleo resin
  b) Gum resin
  c) Oleo gum resin
  d) Glyco resin(gluco resin)
  e) Balsam
- Resins commonly used in pharmacy are derived from natural sources and almost are plant products except Shellac, which an insect secretion

A) OLEO RESIN

- When resins occur in combination with volatile oils, the mixture is called **oleoresin.**
- Volatile oil is removed by **steam distillation**
- E.g. Turpentine, ginger, capsicum, male fern, copaiba etc

B) GUM RESIN

- When resins occur in combination with gums
- Dissolving the associated **gums in water** can purify these resins
- E.g. Asafoetida, gamboge,

C) OLEO-GUM RESIN

- Resins associated with both gums and volatile oils
- Volatile oil is removed by steam distillation while dissolving in water separates gum
- E.g. Myrrh and ipomoea

D) GLYCORESINS (GLUCORESIN)

- Resins found in combination with glycosides
- On hydrolysis produce sugars and complex resins as aglycones
- E.g. Resins which occur in ipomoea, jalap, and podophyllum

E) BALSAMS

- Resinous substances containing high proportions of balsamic acid (benzoic acid, cinnamic acid etc) either in free or in combination or their esters.
- E.g. Tolu balsam contains 35 to 50% of balsamic acid, chiefly benzoic and cinnamic acid
- Benzoin, Peru balsam, and storax are examples of balsams

CHEMICAL COMPOSITION OF RESINS

The 3 important groups of compound in resins are the resin ester, resin acid and resenes.

1.Resin ester:

There are 2 main kinds of resin alcohols in combination with these acids.

- **Resinols:**
  - e.g: storesinol.
- **Resinotannols:**
  - e.g: The resinotannol in Tolu balsam is called Toluresinotannol.

2. Resin acid:

- e.g: Myrrh.

3. Resenes:

These are very stable neutral compounds. They are insoluble in acid, alkali & do not form any salts. They are oxygenated compounds, but are not affected either by alkalies or acids. They are immune to oxidizing agents and variant climatic condition. e.g: Dracoresene (Dragon`s Blood).

Chemical constituents

- Chemically resins are amorphous products with a complex chemical nature.
- Are complex mixtures of the following compounds

1. Resin acids
2. Resin alcohols
3. Resenes

2. Resinols

- High molecular weight non phenolic alcohols
- Do not give tannin reaction with iron salts
- Example: Benzoresinol from benzoin

3. Resenes

- Complex neutral substances
- Do not respond to any chemical reactions
- Insoluble in alkalis and acids
- Do not form salts or esters
- Resist hydrolysis by alkalis
- Example: Alban and fluvial from Gutta percha

Preparation of resins

- *Two general classes of resinous substances are recognized and this classification is based on the method used in preparing them:*

1. **Natural resins**, occur as exudates from plants, produced normally or as result of pathogenic conditions, as for example by artificial punctures e.g. mastic; or deep cuts in the wood of the plant e.g. turpentine, or by hammering and scorching, e.g. balsam of Peru.

2. **Prepared resins**; are obtained by different methods. The drug containing ***resins*** is powdered and extracted with alcohol till exhaustion. The Concentrated alcoholic extract is either evaporated, or poured into water and the precipitated resin is collected, washed and carefully dried.

   In the preparation of ***oleoresins***; ether or acetone having lower boiling point are used. The volatile oil portion is removed through distillation.

   When the resin occurs associated with gum (***gum-resins***), the resin is extracted with alcohol leaving the gum insoluble.

Isolation of resins

- Pharmaceutical resins are obtained from the plants and animals by one of the following methods.
  1. By extraction with alcohol and precipitation with water, e.g. jalap, podophyllum, ipomoea etc.
  2. By distillation for separation of oil, e.g. copoiba, colophony etc.
  3. By heating the plant part, e.g. guaiacum.
  4. As plant exudates by incisions, e.g. myrrh, asafoetida, balsams etc.
  5. By collecting fossil resins, e.g. copal, kauri etc.
  6. By processing the encrustations i.e. shellac.

Identification tests:

- Dissolve 0.1 gm of powdered drug in 10 ml acetic anhydride in a dry test tube. Add a drop of concentrated sulphuric acid to the cold solution. Purple colour appears which quickly changes to dark violet on standing.
- Dissolve powdered drug in petroleum ether. To 5 ml of filtrate, add 10 ml of dilute solution of copper acetate. Petroleum ether layer develops green colour due to formation of abietic acid salt with copper.

MEDICINAL USES OF RESINS

- Resin containing drugs posses many pharmacological activities
- Purgative - Podophyllum
- Cathartic - Gamboge,ipomoea
- Sedative - Cannabis
- Counterirritant - Capsicum
- Expectorant - Storax, benzoin

- Externally, resins are used as **mild antiseptics** in the form of cerates, ointments and plasters

➢ *Drugs containing resins*

1. *Colophony*
2. podopyllum
3. Jalap
4. Mastic
5. Cannabis

Colophony

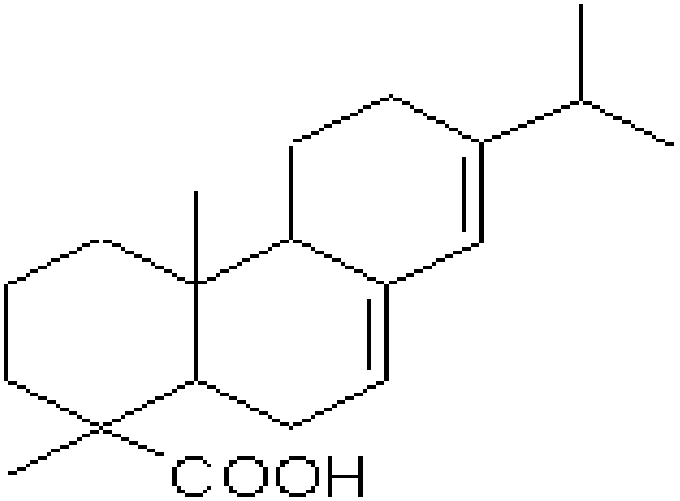
- **Synonyms:** Rosin, Rosina, Amber-resin Pine resin
- **Biological source:** Colophony is the residue left after the distillation of the oil of turpentine from the crude oleo-resin obtained from various species of Pinus
- **Family:** Pinaceae
- **Geographical source**: It is prepared in North America, Northern Europe, India and Pakistan.
- **Description:**
- Colour - Pale yellow to amber coloured fragments
- Odour - Faint and terebinthinate.
- Taste - Characteristic
- Fracture - Transparent, hard, shiny, sharp, brittle fractures at room temperature.

  Burns with smoky flame
- Solubility - Insoluble in water.

  Freely soluble in alcohol, benzene, glacial acetic acid, carbon disulphide, and alkali solutions

➢ Chemical constituents:

- Colophony contains mainly resin acids which are unsaturated and hence reactive.
- About 90% of this is abietic acid.
- The drug also contains esters of oleic acid and resin acids.
- 90% of the resin acids are isomeric α-,β, and γ-abietic acid and 10% is a mixture of dihydroabietic acid and dehydroabietic acid
- It also contains 0.5% volatile oil, 5 to 6% resenes.



Abietic acid

- **Uses:**
  - It is used as a stimulant and diuretic.
  - It is ingredient of certain plaster masses and colloidions.
  - It is also used in the preparation of ointments.
  - Industrially it is used in the preparation of varnishes, insulating compounds, soaps, printing inks, paper sizing.
  - In the preparation of floorings, soldering compounds, mastics and pressure sensitive, adhesives.

PODOPHYLLUM

- **SYNONYMS**: *Podophyllum radix*, Himalayan May-apple.
- **BIOLOGICAL SOURCE**: Indian Podophyllum consists of dried rhizomes and roots of Podophyllum hexandrumor P. emodi. American podophyllum consists of dries rhizomes and roots of P. peltatum.
- **FAMILY**: Berberidaceae.
- **GEOGRAPHICAL SOURCE**: *Podophyllum hexandrum* is found in India, China, and the Himalayas and hence known as Indian podophyllum, while *P. peltatum* (May apple or American mandrake) comes from North America and is known as American podophyllum.
- Plants are collected from the wild.

PODOPHYLLUM

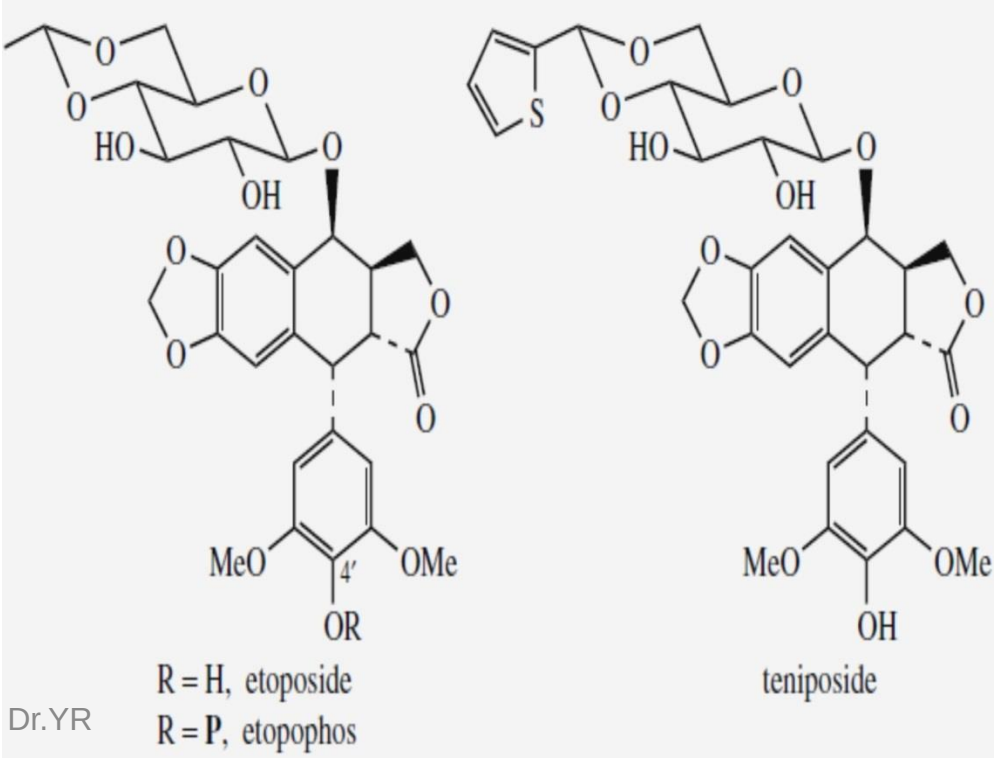
- **MACROSCOPICAL CHARACTERS**:
  - Length : 3-8 cm; Thickness: 1-2cm.
  - Shape : Sub-cylindrical with irregular nodes, tortuous, dorsi -ventrally flattened.
  - Surface :
    - Rhizome: 3-4 circular depressed stem-scars.
    - Roots: longitudinally wrinkled, nearly straight, curved or tortuous.
  - Colour : Earthy brown.
  - Odour : Slight & characteristic.
  - Fracture : Short & brittle.
  - Taste : Bitter & acrid.

Chemical constituents

- The *roots contain cytotoxic lignans* and their glucosides,*P. hexandrum* containing about 5%, and *P. peltatum* about 1%.
- Indian podophyllum yields about 6–12% of resin containing 50–60% lignans, and American podophyllum 2–8% of resin containing 14–18% *lignans.*
- *The lignan constituents of the two roots are the same, but the proportions are markedly different*.

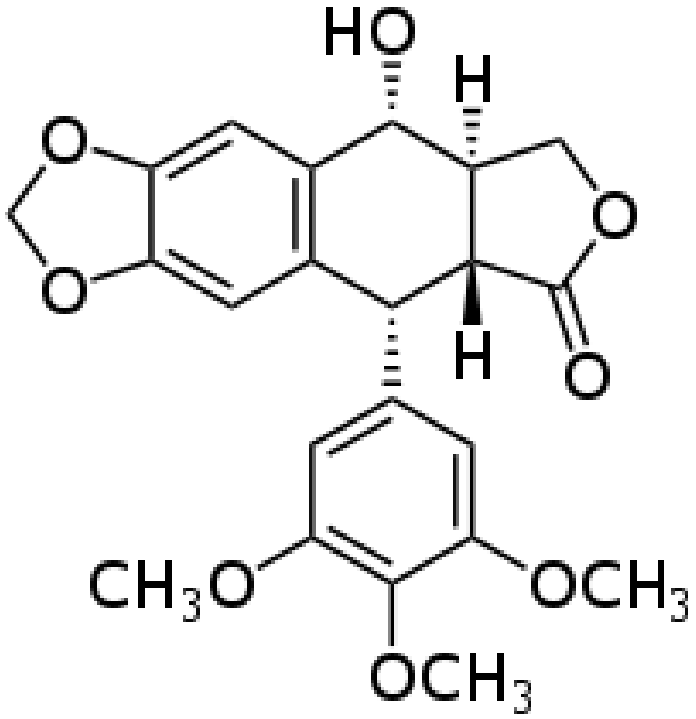
- The semi-synthetic derivatives **etoposide** and **teniposide** are manufactured from natural podophyllotoxin, and
  - Are excellent antitumour agents.

- Etoposide is a very effective anticancer agent, and is used in the treatment of small cell lung cancer, testicular cancer and lymphomas, usually in combination therapies with other anticancer drugs.
- It may be given orally or intravenously.



Dr.YR

- **CHEMICAL CONSTITUENTS:**
  - 7-15% Resin known as podophyllin.
  - Podophyllin: Podophyllotoxin (present prominently in Indian variety); αand β-peltatins (present in American variety only).
  - Podophyllin is a lignan chemically i.e. a substance whose chemical structure is based on 2,3-dibenzyl butane.
  - Flavonoid: Astragalin.
  - Others: Quercetin, kaempferol, essential oils.
  - Etoposide and teniposide are semi-synthetic derivatives.



Podophyllotoxin

- **IDENTIFICATION TESTS**:
  - Macerate 0.5g of the drug with 10ml of alcohol and filter. To the filtrate, add 0.5ml strong copper acetate solution, brown precipitate is produced.
- **USES**:–
  - Cytotoxic effect: Treats veneral and other warts, Treats some forms of cancer. Cytotoxic action by bringing mitotic arrest.
  - Purgative, cholagogue and bitter tonic.
  - Treatment of chronic hepatitis.

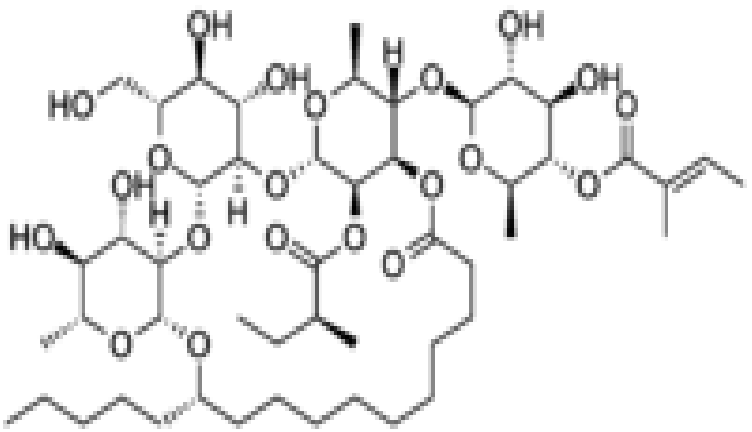
JALAP

- **SYNONYMS**: Mexican Jalap, Jalap radix.
- **BIOLOGICAL SOURCE**: Jalap consists of dried tubercles of *Ipomoea purga* Hayne.
- **FAMILY**: Convolvulaceae.
- **GEOGRAPHICAL SOURCE**: Jamaica, South America, India.
- **CULTIVATION & COLLECTION**:
  - Perennial herb with horizontal runner, nodes and internodes.
  - Nodes characterized by adventitious roots.
  - Has tuberous roots, which swell due to food storage and get converted to tubercles.
  - Collected in May after rainy season.
  - Collected by digging.
  - Dried by wood-fire, as external environment not suitable for drying. As a result drug acquires smoky odour.

JALAP

- **MACROSCOPICAL CHARACTERS**:
  - General appearance: Hard, heavy roots-entire or with slits.
  - Shape : Irregularly oblong, irregular, cylindrical, fusiform or napiform.
  - Surface : Longitudinally wrinkled with transverse lenticels. Sometimes slits are seen in large pieces.
  - Fractures surface: Resinous and greyish.
    - Inside the fractured surface, within the cambium, numerous dark and irregular curved lines seen. These lines are abnormal vascular bundles.
  - Colour : Dark brown.
  - Odour : Smoky.
  - Taste : First sweet, later acrid.

- **CHEMICAL CONSTITUENTS**:
  - 8 to 18% Resin: Ether soluble portion and Ether insoluble portion.
  - Ether soluble portion:
    - Main active constituent of jalap;
    - Known as Jalapin (scammonin).



Jalapin (scammonin)

- Ether insoluble portion:
  - Known as Convolvulin;
  - Convolvulin, on hydrolysis, yields 74% rhamnoconvolvulic acid, 9% tiglic acid, 7% exogonic acid.
- Other components:
  - ipurganol, methyl aesculetin, palmitic and stearic acids.
  - volatile oil, starch, gum and sugar.Exogonic acid.
  - Exogonic acid

JALAP

- **IDENTIFICATION TESTS**:
  - Transversely cut surface and powder appear pale-pinkish in UV light.
- **USES:**
  - Powerful cathartic. Acts in presence of bile. Mechanism of action is parallel to that of ricinoleic acid of castor oil.
  - Extensively used either as standardised powder, as jalap resin or as jalapin.

Mastic

- **Synonyms**: mastich/chios tear
- **Biological source**:
  - Mastic is a resin obtained from the bark of mastic tree *pistacia lentiscus*
- Family: Anacardiaceae
- **Geographical source**:
  - It is native throughout the Mediterranean region, from Morocco, southern France, Turkey, Iraq and Iran. It is also native to the Canary Islands.
- **Description**: Occurs in yellow or greenish yellow round about 3mm diameter

- Chemical constituents

Mastic contains about 90%resins that are:-

1. Alpha resin(mastichic acid)

   Soluble in alcohol

2. Beta resin (maciticin)

   Insoluble in alcohol

- **Uses:**
  - Pharmaceutically as antibacterial and anti fungal, anti ulcer, preparation of mastic paints, microscopical mountant.
  - Non-pharmaceutically(commercially): used in liquorice-cream chewing gums, cosmotics such as lotions, shampos, perfumes and varnish.
  - A high-grade construction adhesive commonly used to bond ceiling, wall, and floor tiles, plywood panels, concrete, asphalt, leather and fabric.
  - A waterproof, putty-like paste used in building as a joint-sealer or filler

cannabis

- **Synonyms**: Indian hemp, Ganja, Marihuana
- **Biological source**: Cannabis consists of dried flowering tops of the cultivated female plants of *Cannabis sativa* Linn.
- **Family**: cannabinaceae.
- **Geographical source**:
  - It is indigenous to India, Commercially produced in Mexico and Africa.
- **Description**:

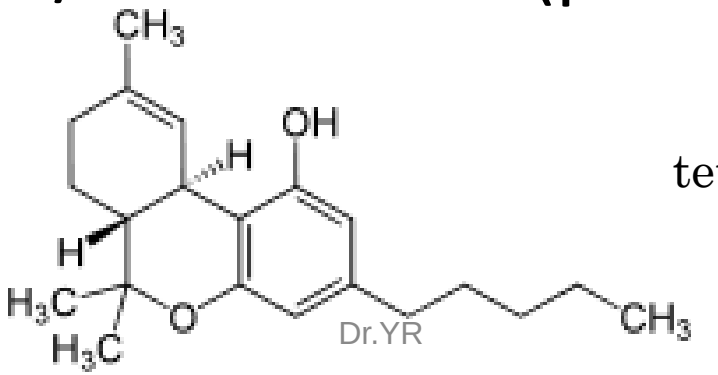
Colour	-	Dull green leaves and after processing it is black to dark brown
Odour	-	Strong, characteristic and narcotic
Taste	-	sweet cloying smell

- When collected and compressed it forms hard blocks

- **Method of preparing herbal drug extract**
  - Decarboxylation
  - Extraction with liquid CO_2
  - Precipitation with alcohol
  - Filtration
  - Evaporation
- **Narcotic products of cannabis:**
  - Bhang or Siddhi
  - Ganja
  - Charas

Chemical constituents

- 15 to 20% of resin.
- The major active principle of resin is 1-3-4 trans tetra-hydrocannabinol.
- The resin also contains cannabinol, cannabidiol, cannabidiolic acid, cannabichromene and cannabigerol.
- It also contains trigonelline and choline, essential oils/volatile oils(pinene, myrcene, ocimene).



tetrahydrocannabinol

- **Uses:**
  - It is a narcotic, sedative and analgesic.
  - It has psychotropic properties due to tetrahydrocannabinol.
  - It causes intoxication, euphoria and later mental disturbances.

Examples of drugs containing oleoresin

1. Ginger
2. Capsicum

GINGER

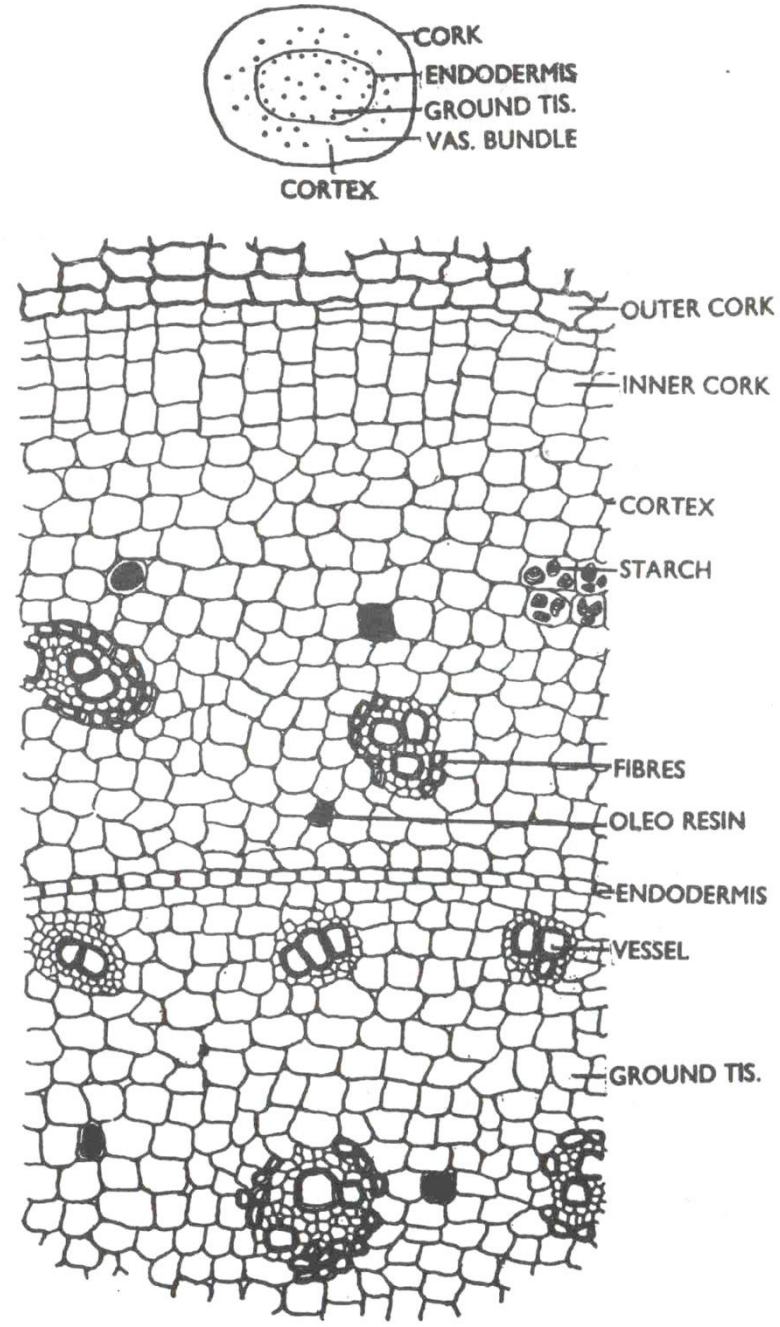
- **SYNONYMS**: zingiber.
- **BIOLOGICAL SOURCE**: Ginger consists of whole or cut, dried scrapped or unscrapped rhizomes of *Zingiber officinale* Roscoe.
- **Family**: Zingiberaceae
  - It contains 0.8% of total gingerols on dried basis.
- **GEOGRAPHICAL SOURCE**:
  - Native to South-east Asia
  - Cultivated in Caribbean islands, Africa, Australia, Mauritius, Jamaica, Taiwan and India.

MACROSCOPICAL CHARACTERS:

- **General appearance**: Sympodial branching, horizontal rhizome.
- **Size**:
  - Length: 5 to 15cm; Width: 3 to 6cm; Thickness: 0.5 to 1.5cm.
- **Shape** : Laterally compressed, bearing short flat, ovate & oblique branches on the upper side, with bud at the apex.
- **Surface** : Longitudinally striated with occasional projecting fibres.
- **Fracture** : Short, starchy, fibrous.
- **Fractured surface** : Shows a narrow bark, a well marked endodermis & a wide stele, showing numerous scattered greyish points (fibrovascular bundles) & smaller yellowish points (secretion cells).
- **Color** **:** Buff.
- **Odour** : Agreeable & aromatic.
- **Taste** : Agreeable & pungent.

MICROSCOPY:

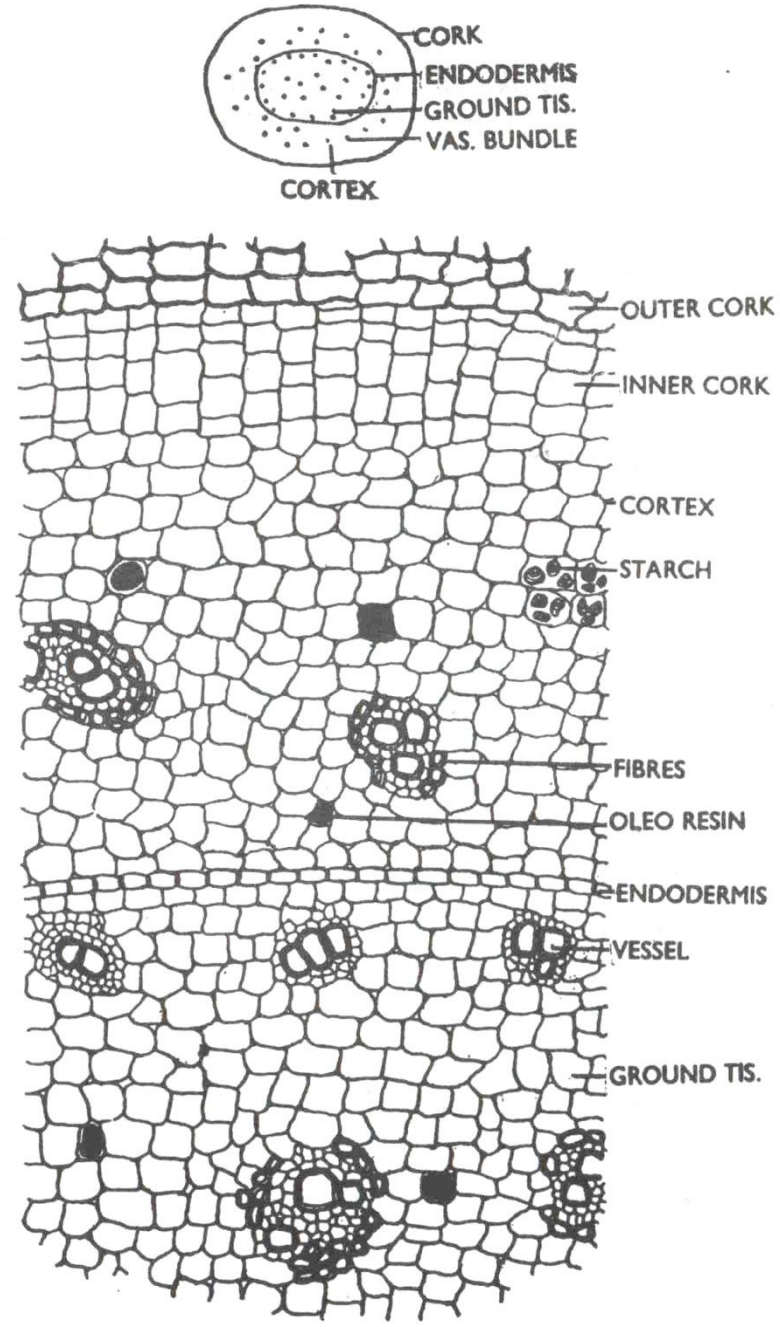
- **Cork**: Outer zone consists of irregularly arranged cells.
  - Inner zoneconsists of cells arranged in radial rows.
  - Cork absent in Jamaica Ginger.
- **Phellogen**: Indistinct.
- **Cortex**: Thin-walled cellulosic rounded parenchyma with intercellular spaces.
  - These contain simple, ovate or sac shaped starch grains with hilum at the pointed end.
  - Contains closed collateral fibrovascular bundles.
  - Some cells contain yellow brown oleo resin cells.



Ginger Rhizome. Diagrammatic and an enlarged portion of T. S. (X 100)

MICROSCOPY:

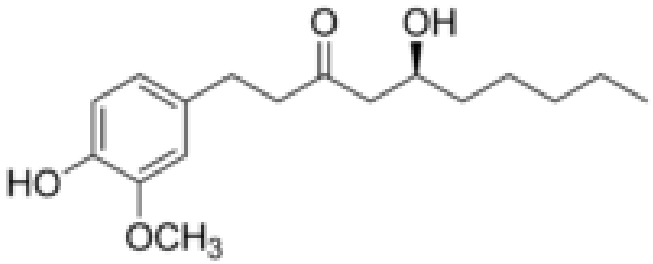
- **Endodermis:** distinct.
  - Consists of tangentially elongated cells containing suberin in radial walls.
  - Starch is absent.
- **Stele:** Below endodermis is a ring of vascular bundles without fibres.
  - The remaining tissue contains fibrovascular bundles, starch and oleo-resin cells similar to cortex.



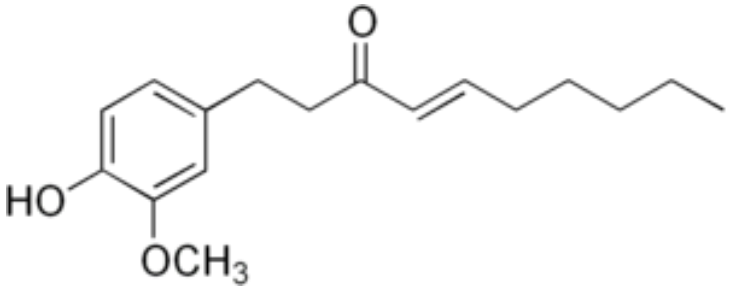
Ginger Rhizome. Diagrammatic and an enlarged portion of T. S. (X 100)

CHEMICAL CONSTITUENTS:

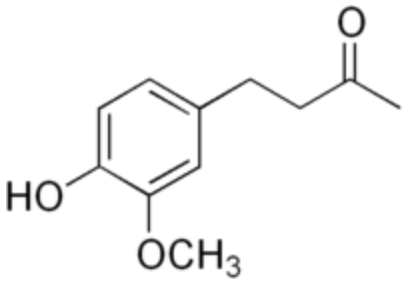
- 5 to 8% resinous mass.
- 1-2% volatile oil: Responsible for aromatic odor of drug.
  - Gingerol-Responsible for pungency of the drug.
  - Sesquiterpene hydrocarbon-α-zingiberol, α-farnesene, α-bisabolene, α-sesquiphellandrene.
  - Less pungent components: Gingerone; Shogoal.
  - Shogaol: Not present in fresh rhizome; Formed by dehydration of gingerol.



Gingerol



Shogaol



Zingerone

- **USES**:
  - Stomachic, an aromatic, carminative & a stimulant.
  - Flavoring agent.
  - Effective in motion sickness.
  - Positive inotropic effect.
  - Spasmolytic.
  - Effective in sore throat, cold, cough & in asthma.
- **ADULTERANTS**:
  - Adulterated by addition of exhausted ginger. It can be detected by determination of water soluble ash, volatile oil content & alcohol and water soluble extractives.

- The pungency of gingerol is lost by boiling with 2% potassium hydroxide.
- Boiling with baryta water decomposes it with formation of zingerone and aliphatic aldehydes.
- The pungency of zingerone is also destroyed by prolonged contact with %5 sodium hydroxide.
- Shogaols are dehydration products of gingerols.
- Ginger essences or extracts are sometimes "fortified" by the addition of capsicum.
- Addition of KOH will not destroy the pungency of capsicum.

Medicinal Uses

- Ginger is an excellent natural remedy for nausea, motion sickness, morning sickness and general stomach upset due to its carminative effect that helps break up and expel intestinal gas.
- Ginger tea has been recommended to alleviate nausea in chemotherapy patients
- Some studies show ginger may also help prevent certain forms of cancer due to its antioxidant properties

CAPSICUM

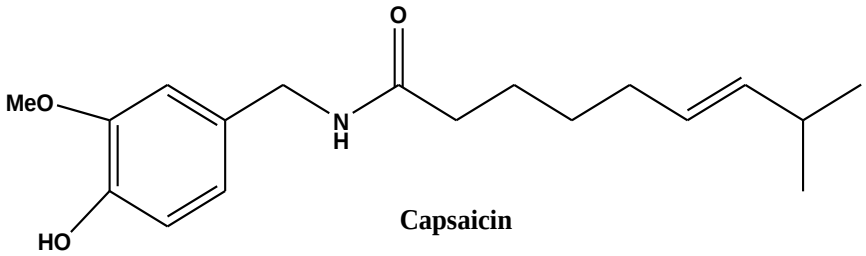
- **SYNONYMS**: Chillies, Cayenne pepper.
- **BIOLOGICAL SOURCE**:
  - Capsicum consists of dried ripe fruits of *Capsicum annum* Linn. var minimum.
  - Capsicum or cayenne pepper is the dried ripe fruit of Capsicum frutenscens, known in commerce African chilies
  - It contains not less than 12% of non-volatile ether soluble extractives.
- **FAMILY**: Solanaceae.
- **GEOGRAPHICAL SOURCE**: All tropical countries-E. Africa, W. Africa and India.

CAPSICUM

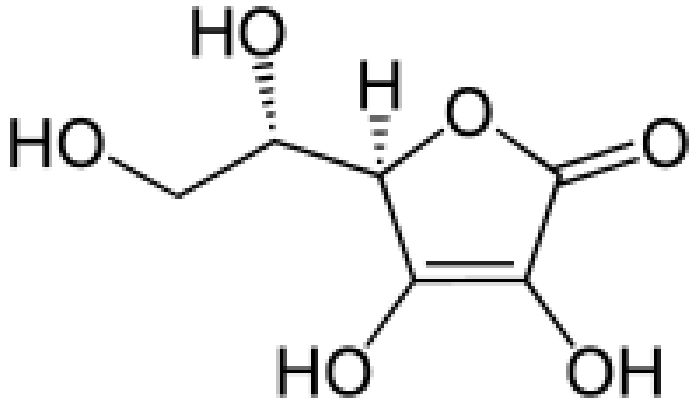
- **MACROSCOPICAL CHARACTERS**:
  - Size: Length-5 to 12cm long; 2 to 4cm wide.
  - Shape: wide & oblong to conical and flattened.
  - Surface: Shriveled, glabrous and leathery.
  - Calyx: 5 toothed, firm and conspicuous.
  - Pedicel: Prominent.
  - Seeds: 10 to 20 brownish yellow flat seeds, about 3 to 4mm long.
  - Color: Dull orange-red to brownish-red.
  - Odour: characteristic.
  - Taste: Pungent.
  - Fruits are divided into two or three loculi.

CHEMICAL CONSTITUENTS

- 0.5 to 0.9% colourless, crystalline and pungent principle known as Capsaicin(amide).
- 0.1 to 0.5% Ascorbic acid.
- Capsainthin and Carotene-pigments responsible for red color. It also contains volatile oils (1.5%), carotenoids, Fixed oils, proteins, a liquid alkaloid &
- Flavones: Eriodictoyland hesperidin.
- The aromatic portion of capsaicin is derived from phenylalanine through ferulic acid and vanillin.



Capsaicin



Ascorbic acid

- **IDENTIFICATION TESTS**:
  - Treating with KOH does not destroy the pungency of capsaicin but The pungency of capsicum is destroyed by boiling it with potassium permanganate.
- **USES**:
  - Used in culinary preparations.
  - Carminative, appetizer and a stomachic.
  - Externally, used as counter-irritant: in rheumatism, inflammation, lumbago, neuralgia.
  - Used in spices.

Examples of drugs containing oleo gum resin

1. **Myrrh**
2. **Asafoetida**

MYRRH

- **SYNONYMS**: Myrrha, Hirabol, Bol.
- **BIOLOGICAL SOURCE**: Myyrh is an oleo-gum-resin obtained from Commiphora molmolEngler and from other Commiphoraspecies.
- **FAMILY**: Burseraceae.
- **GEOGRAPHICAL SOURCE**: North-east Africa, Southern Arabia.
- **COLLECTION**:
  - The tree is small, about 3mt in height.
  - The phloem contains schizogenous ducts and lysigenous cavities which are filled with yellowish granular resinous liquid.

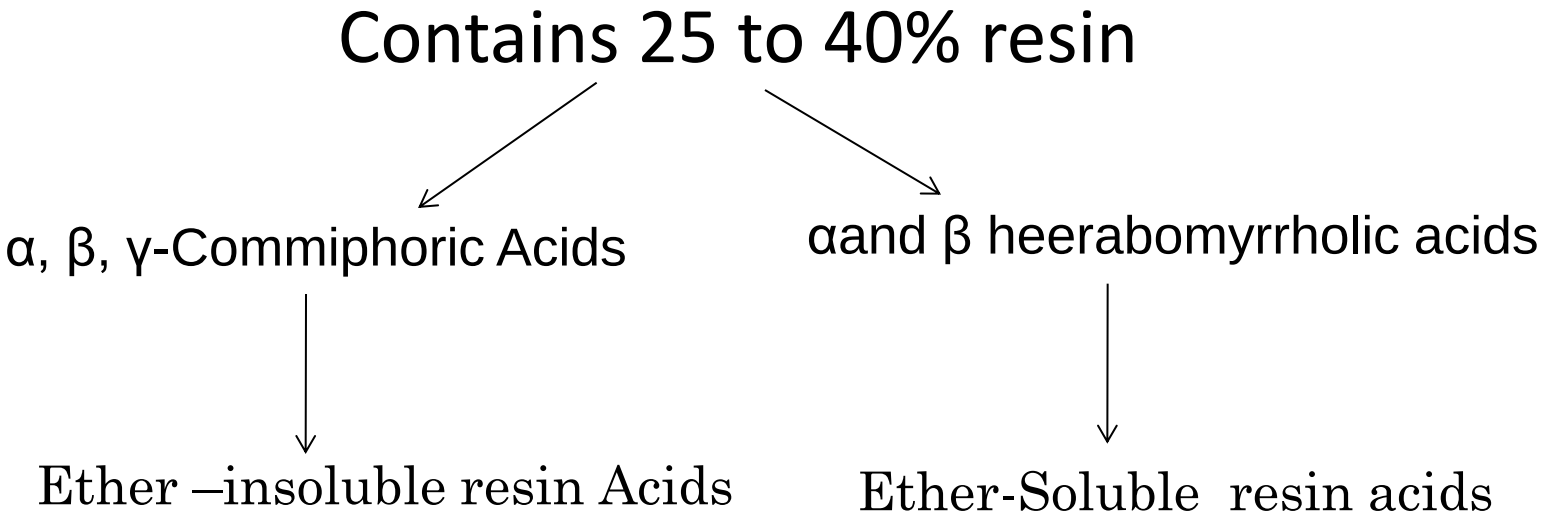
MYRRH

➢MACROSCOPICAL CHARACTERS:

- Size : About 1.5 to 3.0cm in diameter.
- Shape : Found in the form of rounded or irregular tears.
- Color : Externally, it is reddish-brown;
  Internally, brown.
- Odour : Aromatic and agreeable.
- Taste : Aromatic, bitter and acrid.
- Surface : Rough.
- Fractured surface: Waxy, granular and oily with whitish marks.

MYRRH

- **CHEMICAL CONSTITUENTS:**



- 10% yellowish thick volatile oil-terpenes, cuminic aldehyde, eugenol.
- 60% gum. Gum is associated with oxidase enzyme.
- A bitter principle.

MYRRH

➢IDENTIFICATION TESTS:

- 0.1g substance + 0.5g sand ⟶ Triturated with solvent ether ⟶ Filter ⟶ Evaporate ⟶ The thin film formed gives reddish color on contact with bromine vapors and purplish with nitric acid.
- When triturated with water, it forms yellowish brown emulsion.

➢USES:

- Stimulant and an antiseptic; Protective.
- Astringent to the mucous membrane and hence its tincture is used in the mouth-washes and gargles.

➢ADULTERANTS:

- Substituted by several species like Arabian Myrrh, Yemen Myrrh, etc. both of them are less fragrant and less aromatic.
- In India, myrrh is substituted b Balsamodendron mukul, known as Indian bdellium.

ASAFOETIDA

- **SYNONYMS**: Devil's dung,Gum asafoetida.
- **BIOLOGICAL SOURCE**: Asafoetida is the oleo-gum-resin obtained by incision from the rhizomes and roots of Ferula foetida Regel, Ferula rubricaulisRegel and other species of Ferula.
- **FAMILY**: Umbelliferae.
- **GEOGRAPHICAL SOURCE**: Distributed from Mediterranean region to Central Asia, specially Iran and Afhanistan. Also found in India (Kashmir).

ASAFOETIDA

➢ MACROSCOPICAL CHARACTERS:

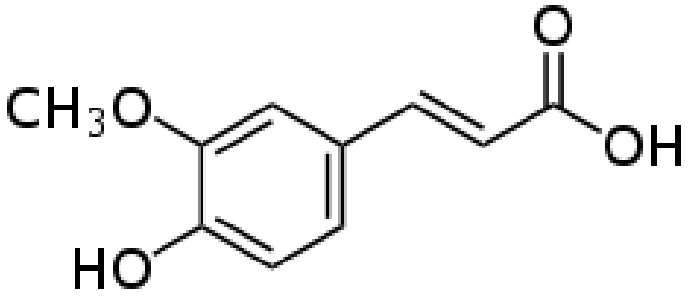
- ➢ Size : Tears are 0.5 to 3cm in diameter.
- ➢ Shape : Occurs in 2 different forms i.e. tears and masses.
- ➢ Tears rounded or flattened.
- ➢ Fresh tears are tough but when dried they become hard and brittle.
- ➢ Mass is agglutinated and mixed with root fragment, foreign material and other impurities.
- ➢ Colour : Yellowish-white changing to reddish-brown.

  Internally milky whitish-yellow, translucent or opaque.
- ➢ Odour :Intense, persistent, penetrating and alliaceous.
- ➢ Taste : Bitter, alliaceous and acrid.

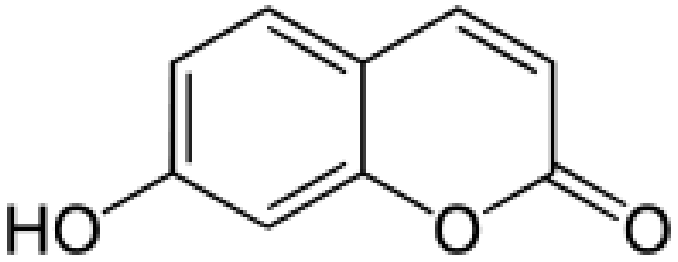
ASAFOETIDA

CHEMICAL CONSTITUENTS:

- 40 to 65% resin-Chiefly asareisontannol in the free or combined form with ferulic acid.
- Ferulic acid, on treatment with HCl, is converted into umbellic acid, which loses water to form Umbelliferone.
- Umbelliferone is not present in the free form.
- Oil of asafoetida is obtained by steam distillation of the oleo-gum-resin.
- Chief constituents of oil: secondary butyl propanyl disulphide.



Ferulic acid



Umbelliferone

ASAFOETIDA

- **IDENTIFICATION TESTS**:
  - Umbelliferone Test: 0.5g drug +sand+ Triturate + 5ml HCl+Water ⟶ Filter ⟶ Filterate + equal volume of Ammonia ⟶ Blue fluorescence, due to presence of umbelliferone.
  - Fractured surface + Sulphuric acid ⟶ Reddish brown color.
  - When triturated with water, it forms the yellowish orange emulsion.
- **USES:**
  - Carminative and in intestinal flatulence.
  - Nervine stimulant.
  - Flavoring agent.
- **ADULTERANTS:**
  - Adulterated with gum arabic, rosin, gypsum, red clay, chalk and barley or wheat flour.

Balsams

- Balsams are resinous mixtures that contain large proportions of cinnamic acid, benzoic acid or both or esters of these acids.
- The term “balsam” is often wrongly applied to oleoresins and should be reserved for such substances as *balsam of Peru*, *balsam of Tolu* and *storax*, which contain a high proportion of aromatic balsamic acids.

BALSAMS

- Balsams are naturally occurring resinous mixtures, which contain high proportion of aromatic balsamic acids i.e. BENZOIC ACID and CINNAMIC ACID and their esters.
- Balsams containing free acids are partially soluble in hot water.
- Some important balsam containing crude drugs are:-

- Balsam of Peru,
- Balsam of Tolu,
- Benzoin and Storax.

TOLU BALSAM

- **Synonym:** Balsamum tolutanum
- Tolu balsam is obtained from the incisions made on the tree trunk of *Myroxylon balsamum*.
- The fresh balsam is soft, yellowish resinous mass which upon storage becomes brownish brittle mass.
- It has pleasant odor and aromatic balsamic taste.
- It is soluble in alcohol; it gives green color with $FeCl_3$ solution and it produces the odor of benzaldehyde due to the oxidation of cinnamic acid.
- It contains about 80% of resin alcohols in combination with the balsamic acids.
- It contains free balsamic acids, benzyl benzoate, benzyl cinnamate, ferulic acid, vanillin and eugenol.
- Balsam of Tolu is used as an important ingredient of cough preparations.

PERU BALSAM

- **Synonym:** Balsamum peruviamum
- Peru balsam is derived from the injuries in the trunk of *Myroxylon pererae*.
- It is a blackish or dark-brown, viscous, oily liquid in the form of thin layer.
- It has pleasant aromatic vanilla like odor and acrid bitter taste.
- Peru balsam contains esters of the balsamic acids (benzyl benzoate, benzyl cinnamate, and cinnamyl cinnamate) and peruresinotannol.
- Balsam of Peru is possesses local protectant and rubefacient properties.
- It is used as an antiseptic and parasiticide in certain skin diseases such as scabies.
- It is also used for the treatment of haemorrhoids due to its astringent properties and as an ingredient in paraffin gauge dressings.

BENZOIN

- **SYNONYMS**: Sumatra Benzoin, Loban.
- **BIOLOGICAL SOURCE**: Benzoin is the balsamic resin obtained from *Styrax benzoin* Dyrand or *Styrax* paralleloneurus Perkins, known in the market as **Sumatra benzoin**; or it may also contain the balsamic resin from Styrax tonkinesis and other species commercially known as **Siam benzoin**.
- **Family**: Styraceae.
  - It should contain not less than 25% of total balsamic acids, with reference to dry alcohol soluble matter.
- **GEOGRAPHICAL SOURCE**:
  - Sumatra Benzoin: South Eastern Asia, chiefly Sumatra.
  - Siam benzoin: Thialand, Vietnam.

COLLECTION & PREPARATION FOR MARKET:

- Benzoin is the pathological resin, secreted in secretory ducts & secretory cells of the tree after injury.
- Fungus also takes part in production of benzoin.

Seeds are sown in rice fields. Rice plants give protection to benzoin plant for 1 yr.

↓

After harvesting rice, trees of benzoin are allowed to grow till they are 7yrs old.

↓

Collection of resin is carried out from trees of 7 to 20yrs.

↓

A triangular gaping wound is made at the base of the tree, exposing the xylem.

BENZOIN

After about a week, yellow, soft, sticky resinous secretion

After about 2 months, solidification & scrapped off.

This 1st secretion is yellow colored & amorphous and not utilized in medicine.

New similar wounds at distance of 4cm above the earlier wounds are made on the stem.

2nd exudation is of the best quality and is known as benzoin almond.

↓

The 2nd exudation is milky white and crystalline.

In an year, process of wounds is carried out 4 times.

↓

3rd and 4th exudations are darker in color and inferior.

A single tree produces approximately 10kg material.

MACROSCOPICAL CHARACTERS:

1. **Sumatra Benzoin**:
   - Occurs in the form of varying sizes of lumps or tears.
   - Fractured surface: uneven
   - Appearance: Mass is embedded in translucent, reddish brown resinous matrix.
   - Color: Greyish brown or grey.
   - Odor: aromatic & characteristic.
   - Taste: sweetish & slightly acrid.
   - When heated, fumes of benzoic acid & cinnamic acid is produced.

2. **Siam Benzoin**:

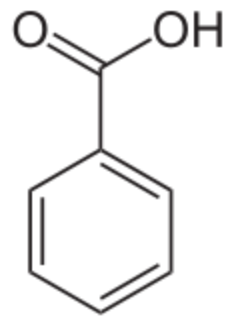
- Color: Yellowish-brown
- Odour: Agreeable & vanilla-like.
- Taste: Sweetish & slightly acrid.
- Occurs as hard & brittle masses.
- When heated, it is softened and becomes plastic.

STANDARDS:

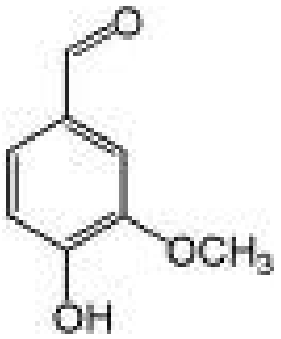
- Benzoic acid content:
  - Not less than 60% (sumatra)
  - Not less than 12% (siam)
- Alcohol soluble matter:
  - Not less than 76% (sumatra)
  - Not less than 90% (siam)
- Loss on drying: Not more than 10%
- Acid insoluble ash: Not more than 10%

➢ CHEMICAL CONSTITUENTS:

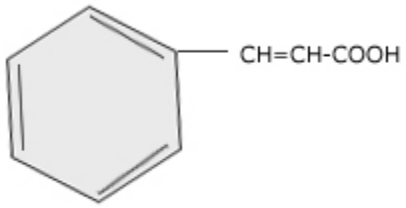
- **Sumatra benzoin**: Free balsamic acids, i.e. Benzoic and cinnamic acids; & esters derived from them.
  - Triterpenoid acids-summaresinolic acid, siaresinolic acid.
- **Siam benzoin**: Mainly-ester coniferyl benzoate (76%).
  - Also styrol, vanillin, phenyl propyl cinnamate.
- Siam benzoin differs from sumatra variety in that it contains insufficient cinnamic acid to give an odor of benzaldehyde, when warmed with $KMnO_4$.



Benzoic acid



Vanillin



CINNAMIC ACID

- **IDENTIFICATION TESTS:**
  - Alcoholic solution of benzoin + Water ⟶ Milky white solution.
  - Heat small qty of benzoin in a test tube covering opening of test tube with a glass plate. Examine the glass plate under microscope. The crystals of cinnamic acid shall be observed.
  - 2.5g benzoin + 10ml ether. Shake well. Pour 2-3ml of the extract in a porcelain dish + 2-3 drops of $H2SO_4$ ⟶ A deep brown colour in Sumatra benzoin & a deep purplish red colour in case of Siam benzoin.
  - 1g benzoin + 4ml $KMnO_4$ ⟶ Odour of benzaldehyde.

TESTS FOR IDENTITY

- Treatment of Siam Benzoin with ferric chloride develops green color while Sumatra benzoin remains unaffected.

  Siam Benzoin +$fecl_3$ =green color

  sumatra Benzoin +$fecl_3$=no change

- Treatment of Sumatra benzoin with dil. H_2SO_4 and $KMnO_4$ with gentle heating evolves benzaldehyde (detected by its sweet odour) indicating the presence of cinnamic acid.

Sumatra benzoin + dil. H_2SO_4 + $KMnO_4$ + heat= benzaldehyde indicating the presence of cinnamic acid.

- **USES:**
  - Internally as expectorant and externally as antiseptic & a protective.
  - Carminative
  - Diuretic
  - In the form of compound tincture of benzoin.
  - In treatment of upper respiratory tract infection.
  - Preferred to retard the rancidity of fats & oils in benzoated lard.
  - Industrially, it is used to fix the odour of incense, soaps, perfumes & several other cosmetics and to mask the taste of pharmaceuticals.

Crude drugs containing Resins

Name of Crude drug	Biological Source	Chemical Constituents	Uses
Podophyllum (Indian apple)	*Podophyllum emodi*, Family: Berberidaceae	Podophyllotoxin, Picropodophyllin Flavones starch	Anticaner agent
Asafoetida (Devil's dong)	*Ferula foetida*, Family: Umbelliferae,	Ferolic acid, Asaresinol Pinene	Stimulant,Carminative
Ginger, (Zingiber)	*Zingiber afficinale*, Family: Zingiberaceae	Zingiberine, Curcumene, Phenyl propanoids	Stimulant, Aromatic, Carminative
Turmeric (Indian saffron)	*Curuma longa* Family: Zingiberaceae	Curcumine -I, II, III, Sugar, Camphene	Condiment, Colouring agent
Capsicum (Red pepper)	*Capsicum annum* Family: solanacea	Capsaicin, Casanthin, Carotin Protein, Ascorbic acid	Stomachic, Stimulant
Benzoin (Loban)	*Styrax benzoin* Family: Styraceae	Siaresinolic acid, Sumaresinolic acid Vanillin, Cinnamic acid	Diuretic, Expectorant, Carminative

Name of Crude drug	Biological Source	Chemical Constituents	Uses
Colophony (Rosina)	*Pinus elliotti*, Family: Pinaceae	Abietic acid Pimaric acid, Sapinic acid	Stimulant, Diuretic
Shellac (Lac)	Lcifer lacca Family: Lacciferidae Order: Hemiptera	Resin, shelloic acid aleuritic acid, kerrolic acid, butolic acid	Sustained release medicaments
Cannabis (Indian hemp)	*Cannabis Sativa*, Family: *Cannabinaceae*	Tetrahydro-cannabinaol, Cannabinaol Trigonelline Choline	Antitussive, Antianxiety,
Guggul	*Commiphora weightii* Family: Burseraceae	Gum resin, steroids , guggulostrones	Antiinflammatory, Hypolipidemic